

A First-Order Conditional Singular Series for Twin-Prime Gaps on the $6N$ Skeleton, and Its Second-Order Residual

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June 1, 2026

Abstract

Part II reported that the twin-prime gap distribution on the $6N \pm 1$ skeleton depends on the factor count $\omega_{>3}(N)$ of the centre—the preference for the gap $6\Delta N = 42$ rises with ω while that for 210 falls—and left the closed-form conditional gap singular series as an open problem. Here we give its explicit first-order form. Writing the unconditional Hardy–Littlewood correlation factor $C_2(d)$ and conditioning on the centre’s factor set through the congruence-lockdown rule of Part II ($q \mid N$ forbids the gap residues $d \equiv \pm 6^{-1} \pmod{q}$, i.e. $q \mid (6d \pm 1)$), we obtain $\mathfrak{S}_{2,\omega}(d) \approx C_2(d) L_\omega(d)$, where $L_\omega(d)$ is the lockdown-allowance of d in the stratum. Tested against the 23,988,173 twin centres of S_{10} , this first-order series reproduces the ω -trends of the gaps 30, 42, 60 and the direction of the collapse of 210. A systematic residual remains at high ω on the gap 210 (observed falls to 0.41 where the first-order series predicts 0.62), stable across S_9 and S_{10} ; we attribute it to the second-order correlation between the factorisations of the two centres N and $N + d$, which the first-order (per-prime independent) series omits. The open problem of Part II is thus advanced from fully open to first-order-solved with a quantified second-order remainder. No claim is made about the infinitude of twin primes or about any prime k -tuple conjecture.

1 The open problem and the first-order series

Part II [2] established, on the $6N \pm 1$ skeleton, that the gap ΔN between consecutive twin centres has a distribution whose shape depends on $\omega_{>3}(N)$, the number of distinct prime factors > 3 of the left centre. The unconditional Hardy–Littlewood theory cannot produce this: its gap singular series depends only on the gap. Part II identified the first-order mechanism—*congruence lockdown*—and posed the closed-form *conditional* singular series $\mathfrak{S}_{2,\omega}(d)$ as an open problem. We give its first-order form and test it.

Derivation (first order). For a gap d , two twin centres N and $N + d$ are involved. Per prime $q > 3$ we condition on whether q divides the centre N :

- $q \mid N$. The left centre is automatically safe ($6N \pm 1 \equiv \pm 1 \not\equiv 0$). By Proposition 1 of Part II the right centre $N + d$ is safe iff $d \not\equiv \pm 6^{-1} \pmod{q}$, equivalently $q \nmid (6d \pm 1)$. So q contributes a factor 0 if $q \mid (6d \pm 1)$ (the gap is locked out), and otherwise removes the usual suppression.
- $q \nmid N$. The usual unconditional factor $(1 - \nu_q(d)/q)/(1 - 2/q)^2$ applies, with $\nu_q(d) = |\{-1, 1, 6d - 1, 6d + 1\} \pmod{q}|$.

Collecting terms, the first-order conditional singular series for a stratum of centres with factor count ω is

$$\mathfrak{S}_{2,\omega}(d) \approx C_2(d) L_\omega(d), \quad L_\omega(d) = \Pr_{N: \omega_{>3}(N)=\omega} \left[q \nmid N \text{ for all } q \mid (6d \pm 1) \right],$$

where $C_2(d) = \prod_{q>3} (1 - \nu_q(d)/q) / (1 - 2/q)^2$ is the unconditional two-endpoint factor and $L_\omega(d)$ is the *lockdown-allowance*: the probability that a stratum- ω centre divides none of the primes that lock d . The admissibility of d is thus governed by the factorisation of $6d \pm 1$: e.g. $6 \cdot 35 \mp 1 = 209 = 11 \times 19$ (so 210 is locked by the common small primes 11, 19), whereas $6 \cdot 7 \mp 1 = 41, 43$ are prime (so 42 is locked only by the rare primes 41, 43). This is the explicit object Part II asked for; it reduces to the unconditional series when averaged over all ω .

2 Confrontation with S_{10} data

We evaluate $L_\omega(d)$ directly from the 23,988,173 twin centres of S_{10} (computing each centre's small-prime factor set), form the normalised prediction $\propto C_2(d)L_\omega(d)$, and compare to the observed relative preference $r(d \mid \omega)$. Figure 1 shows the four representative gaps.

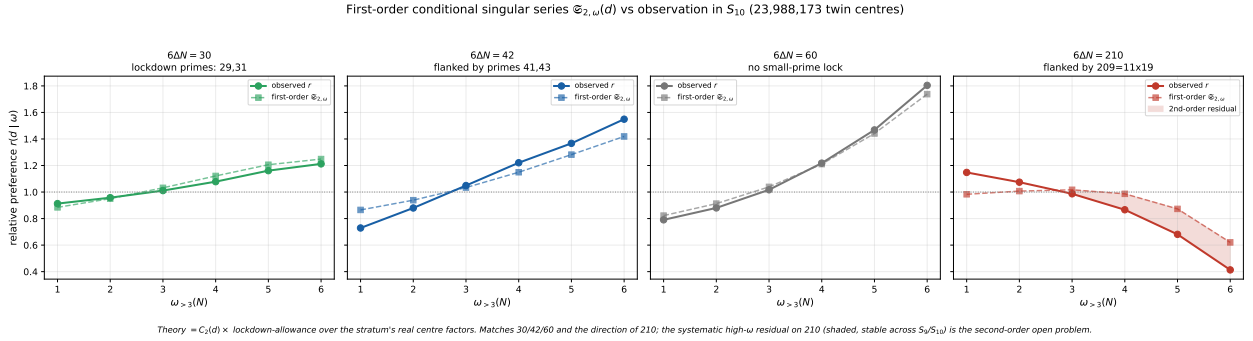


Figure 1: First-order $\mathfrak{S}_{2,\omega}(d)$ (dashed) versus observed $r(d \mid \omega)$ (solid) in S_{10} . The first-order series tracks 30, 42 and 60 in direction and approximate magnitude, and gives the correct direction for 210; the shaded region on 210 is the systematic high- ω residual.

$6\Delta N$	quantity	$\omega=1$	2	3	4	5	6
42	observed	0.73	0.88	1.05	1.22	1.37	1.55
	first-order	0.87	0.94	1.03	1.15	1.28	1.42
210	observed	1.15	1.07	0.99	0.87	0.68	0.41
	first-order	0.98	1.01	1.02	0.99	0.87	0.62

Table 1: Observed vs. first-order $\mathfrak{S}_{2,\omega}$ in S_{10} for the two principal gaps. The series captures 42 closely; for 210 it gives the correct downward direction but under-predicts the high- ω fall, leaving the residual of §3.

The first-order series performs as follows. For $6\Delta N = 30, 42, 60$ it tracks the observed trend in both sign and approximate magnitude across all six strata (residuals within ~ 0.13). For $6\Delta N = 210$ it predicts the correct *direction*—the lockdown by the small primes 11, 19 of 209 makes factor-rich centres increasingly forbidden from this gap—but it *under-predicts* the magnitude of the fall at high ω .

3 The second-order residual

The residual on 210 is systematic, monotone, and stable across scales: at $\omega = 5$ the observed preference is 0.68 against a first-order prediction of 0.87 (residual -0.19) in S_{10} , matching -0.20 in S_9 ; at $\omega = 6$ the observed 0.41 against predicted 0.62 (-0.21). It is not noise.

The first-order series treats the locks from distinct primes as independent and, crucially, conditions only on $q \mid N$ at the *left* centre. It omits the correlation between the factorisations of N and $N+d$: when N is factor-rich, $N+d$ is not independent of it, and for the long gap 210 this correlation further suppresses the joint twin survival beyond the single-centre lockdown. A complete account requires the genuine two-centre conditional singular series,

$$\mathfrak{S}_{2,\omega}^{(2)}(d) = \text{H-L density of } \{6N-1, 6N+1, 6(N+d)-1, 6(N+d)+1\} \text{ conditioned on } \omega_{>3}(N),$$

including the cross terms in the factorisations of the two centres. We have not obtained this in closed form.

Open problem (second order). Derive the two-centre conditional singular series $\mathfrak{S}_{2,\omega}^{(2)}(d)$ and show whether its cross-correlation terms account for the high- ω residual on 210 (observed -0.2 below the first-order series at $\omega = 5, 6$, stable across S_9, S_{10}).

4 Conclusion

We have given the explicit first-order conditional singular series $\mathfrak{S}_{2,\omega}(d) = C_2(d)L_\omega(d)$ that Part II posed as an open problem, derived from the congruence-lockdown rule and verified against 2.4×10^7 twin centres. It reduces to the unconditional Hardy–Littlewood series in the mean, reproduces the ω -dependence of the gaps 30, 42, 60, and predicts the direction of the 210 collapse. The Part II open problem is thereby advanced from fully open to first-order-solved, with a sharply localised second-order remainder: a stable high- ω residual on the long gap 210, attributable to the two-centre factor correlation omitted at first order, and posed here as the next open problem. We make no claim regarding the infinitude of twin primes or any k -tuple conjecture; this is a conditional, factor-resolved refinement of the Hardy–Littlewood gap heuristic, demonstrated empirically and offered with its open remainder to analytic number theory.

Data and code availability. The scan, the first-order series evaluation, and the figure code are openly available [5]; the S_{10} twin count 23,988,173 matches Part I.

References

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